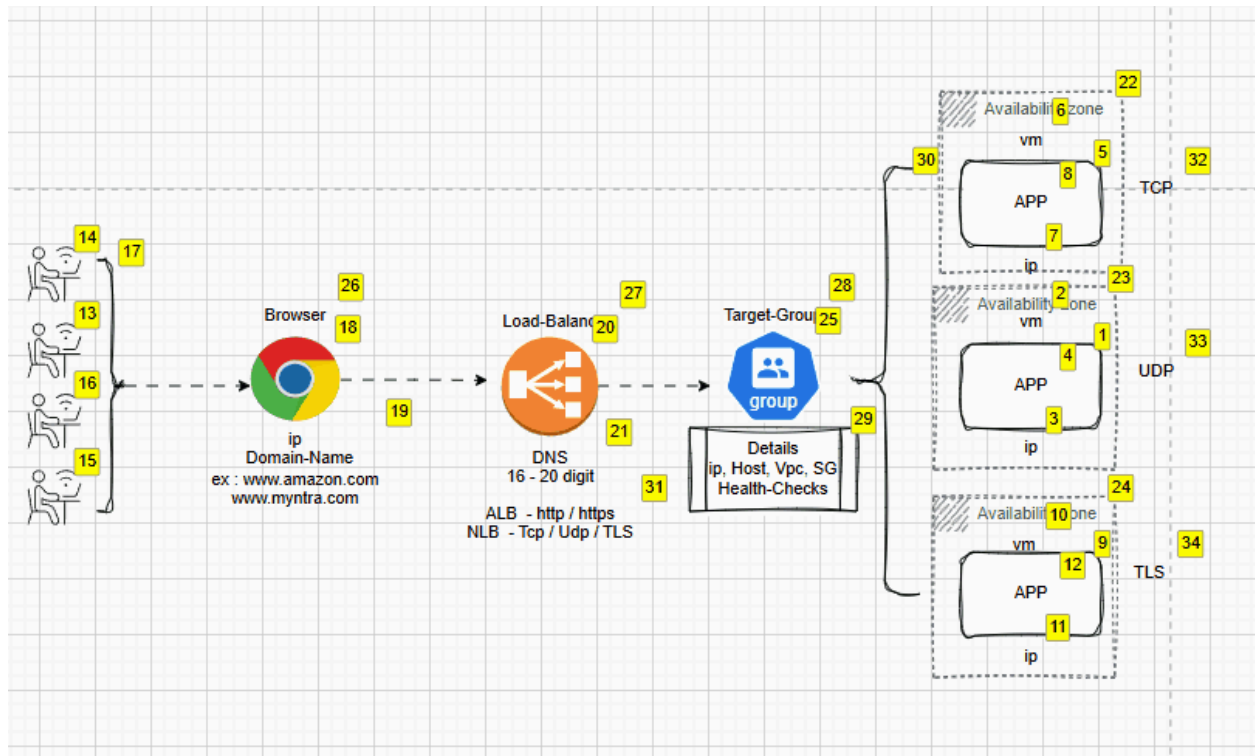


A **Network Load Balancer (NLB)** is a type of load balancer that distributes incoming network (usually TCP/UDP) traffic across multiple backend servers (also called targets) at the transport layer (Layer 4) of the OSI model. It is designed to handle high volumes of traffic with ultra-low latency.



Key Components of a Network Load Balancer

Here are the main components of a Network Load Balancer:

1. Listeners

- **Definition:** A listener checks for connection requests from clients.

- **Details:** It defines the protocol (e.g., TCP, UDP) and the port (e.g., 80, 443) that the NLB uses to listen for incoming traffic.
 - **Example:** A listener on port 80 using TCP.
-

2. Targets

- **Definition:** These are the destination systems where the NLB sends the traffic.
 - **Types:** Targets can be:
 - EC2 instances (in AWS)
 - Containers (e.g., in Kubernetes)
 - IP addresses
 - **Target Groups:** Targets are grouped together into target groups for better management.
-

3. Target Groups

- **Definition:** Logical groupings of backend targets.
- **Functions:**
 - Health checks are defined per group.
 - Routing decisions are made based on group rules.
- **Example:** All web servers might be in one target group.

4. Health Checks

- **Definition:** Automated tests performed by the NLB to check if a target is healthy (able to handle traffic).
- **Parameters:**
 - Protocol (e.g., TCP)
 - Port
 - Thresholds (e.g., 3 failed checks = unhealthy)
- **Purpose:** Ensures traffic is only sent to healthy targets.

5. DNS Name

- **Definition:** The entry point for client traffic, provided by the NLB.
- **Example:** When a client wants to connect, it resolves the NLB's DNS name to the appropriate IP addresses.

6. Elastic IPs / Static IPs

- **Definition:** NLB can be associated with static IPs for predictable access.
- **Use:** Useful for whitelisting or DNS configurations.

7. Cross-Zone Load Balancing (optional)

- **Definition:** Distributes traffic evenly across all targets in all enabled Availability Zones.
 - **Benefit:** Prevents overloading targets in a single zone.
-

How It Works (Flow)

1. A client makes a request to the NLB's DNS name.
 2. The DNS resolves to one or more static IPs assigned to the NLB.
 3. The NLB listener receives the traffic on a specified port and protocol.
 4. The load balancer picks a healthy target from the target group.
 5. The request is forwarded directly (Layer 4 routing) to the target.
 6. The target processes the request and returns the response directly to the client.
-

Advantages of NLB

- High throughput & low latency
 - Supports millions of requests per second
 - Works at Layer 4 (TCP, UDP)
 - Static IP and Elastic IP support
 - Can preserve the client IP for backend logging or security
-

 Comparison: NLB vs. Other Load Balancers

Feature	Network LB (NLB)	Application LB (ALB)	Classic LB (CLB)
Layer	4 (Transport)	7 (Application)	4 & 7
Protocols	TCP, UDP, TLS	HTTP, HTTPS, gRPC	HTTP, HTTPS, TCP
Latency	Very low	Higher than NLB	Medium
Client IP Preservation	Yes	Optional	Yes
